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Comprehensive Question Bank: Unit 5 — Data Analytics

Section A: Multiple-Choice Questions (MCQs)

Topic 5.1: Basic Statistical Concepts (Overview)

1. Data analytics is best defined as:
 - A) The process of designing computer hardware
 - B) The process of examining data to find useful information, patterns, and trends to support decision-making**
 - C) A method of encrypting sensitive information
 - D) A programming language used for web development
 2. Statistics, as described in this unit, is best defined as:
 - A) A branch of biology studying living organisms
 - B) A branch of mathematics that helps us understand and analyze data**
 - C) A method for building physical computer networks
 - D) A type of data visualization software
-

Topic 5.1.1: Measures of Central Tendency

1. Measures of central tendency help identify:
 - A) How spread out a dataset is
 - B) The "center" or typical value in a dataset**
 - C) The total number of values in a dataset
 - D) The probability of an event occurring
2. Which of the following is NOT one of the three main measures of central tendency?
 - A) Mean
 - B) Median
 - C) Mode
 - D) Variance**
3. The mean of a dataset is calculated by:
 - A) Finding the middle value when data is arranged in order
 - B) Adding all the numbers together and dividing the sum by the total number of values**
 - C) Identifying the number that appears most frequently
 - D) Subtracting the smallest value from the largest value

4. What is the mean of the scores 50, 60, 70, 80, and 90?
- A) 60
 - B) 70**
 - C) 80
 - D) 90
5. The median of a dataset is defined as:
- A) The average of all values in the dataset
 - B) The middle value in a dataset when the numbers are arranged in order**
 - C) The value that appears most frequently
 - D) The difference between the highest and lowest values
6. If a dataset has an even number of values, how is the median calculated?
- A) By selecting the larger of the two middle values
 - B) By taking the average of the two middle values**
 - C) By selecting the smaller of the two middle values
 - D) By adding all values and dividing by the total count
7. What is the median of the dataset 50, 60, 70, and 80?
- A) 60
 - B) 65**
 - C) 70
 - D) 75
8. The mode of a dataset is defined as:
- A) The average of all values
 - B) The number that appears most often in a dataset**
 - C) The middle value when data is arranged in order
 - D) The square root of the variance
9. What is the mode of the dataset 50, 60, 70, 70, and 90?
- A) 50
 - B) 60
 - C) 70**
 - D) 90
10. In the dataset 50, 60, 70, 70, 60, and 90, how many modes exist?
- A) Zero
 - B) Two (60 and 70)**
 - C) Three
 - D) One (only 70)
-

Topic 5.1.2: Measures of Dispersion (Variance and Standard Deviation)

1. Measures of dispersion tell us:
 - A) The most frequent value in a dataset
 - B) How spread out or scattered the data is**
 - C) The middle value of a dataset
 - D) The total number of observations in a dataset
2. Which of the following are the two common measures of dispersion discussed in this unit?
 - A) Mean and median
 - B) Variance and standard deviation**
 - C) Mode and probability
 - D) Bar charts and histograms
3. A higher variance indicates that:
 - A) The numbers are closer to the mean
 - B) The numbers are more spread out from the mean**
 - C) The dataset has no mode
 - D) The dataset contains only positive numbers
4. In the variance formula, what does "N" represent?
 - A) The mean of the dataset
 - B) The total number of values in the dataset**
 - C) A single individual value in the dataset
 - D) The standard deviation
5. Based on the worked example in this unit, which class (Class A: 50, 52, 55, 57, 60 or Class B: 30, 45, 55, 75, 90) has a higher variance?
 - A) Class A
 - B) Class B**
 - C) Both classes have equal variance
 - D) Variance cannot be determined from this data
6. What is the mean of Class A's scores (50, 52, 55, 57, 60), as calculated in this unit?
 - A) 50
 - B) 54.8**
 - C) 57
 - D) 60
7. What is the calculated variance of Class A, according to this unit's worked example?
 - A) 4.8
 - B) 12.56**
 - C) 62.8
 - D) 454

8. What is the calculated variance of Class B, according to this unit's worked example?
- A) 12.56
 - B) 59
 - C) 295
 - D) 454**
9. Standard deviation is best described as:
- A) The sum of all values in a dataset
 - B) The square root of the variance**
 - C) The middle value of a dataset
 - D) The most frequently occurring value
10. Why is standard deviation often considered more practical and interpretable than variance?
- A) It is always a smaller number than variance
 - B) It is expressed in the same unit as the original data**
 - C) It does not require any calculations
 - D) It eliminates the need for calculating the mean
11. According to this unit's worked example, what is the approximate standard deviation of Class A?
- A) 12.56
 - B) 3.55**
 - C) 21.26
 - D) 454
12. According to this unit's worked example, what is the approximate standard deviation of Class B?
- A) 3.55
 - B) 12.56
 - C) 21.26**
 - D) 62.8
13. Based on the standard deviation values calculated in this unit, which conclusion is correct?
- A) Class A's scores are more widely scattered than Class B's
 - B) Class A's scores are closely packed around the mean, while Class B's scores are more widely scattered**
 - C) Both classes have identical score distributions
 - D) Standard deviation cannot be used to compare two classes
-

Topic 5.1.3: Introduction to Probability

1. Probability is best defined as:
 - A) The average of all values in a dataset
 - B) The study of how likely an event is to happen**
 - C) The process of cleaning raw data
 - D) A type of data visualization
 2. According to the formula given in this unit, probability is calculated as:
 - A) Total number of outcomes divided by number of favorable outcomes
 - B) Number of favorable outcomes divided by total number of outcomes**
 - C) The mean of all possible outcomes
 - D) The variance of all possible outcomes
 3. When flipping a fair coin, what is the probability of getting heads, according to this unit?
 - A) 25% (or 1/4)
 - B) 50% (or 1/2)**
 - C) 75% (or 3/4)
 - D) 100% (or 1/1)
 4. Which of the following real-world applications of probability is mentioned in this unit?
 - A) Predicting the weather
 - B) Making business decisions
 - C) Predicting outcomes in sports like cricket
 - D) All of the above**
-

Topic 5.2.1: Data Collection Methods

1. Data collection is best defined as:
 - A) The process of visualizing data using charts
 - B) The process of gathering relevant information for a particular purpose**
 - C) The process of calculating variance and standard deviation
 - D) The process of building a linear regression model
2. Which of the following is NOT one of the three data collection methods discussed in this unit?
 - A) Surveys
 - B) Observations
 - C) Experiments
 - D) Clustering**
3. Surveys are best described as:
 - A) Watching subjects in their natural environment without interference
 - B) Asking a predefined set of questions to a sample group, often via online forms,**

phone calls, or face-to-face interviews

- C) Manipulating variables to determine their effect on another variable
 - D) A method used only for numerical data
4. In the grocery store example in this unit, how many customers were surveyed?
- A) 5
 - B) 50**
 - C) 100
 - D) 500
5. Observation, as a data collection method, is best described as:
- A) Asking a predefined set of questions to a group
 - B) Collecting data by watching or monitoring subjects in their natural environment**
 - C) Manipulating variables in a controlled setting
 - D) Estimating missing values using existing data
6. In the restaurant example in this unit, what did the staff member observe over one week?
- A) Customer complaints
 - B) Which tables were most frequently chosen by customers during lunchtime**
 - C) Employee attendance
 - D) Menu pricing changes
7. Experiments, as a data collection method, involve:
- A) Watching subjects without any interference
 - B) Manipulating one or more variables to determine their effect on another variable**
 - C) Asking a predefined set of questions
 - D) Only collecting data from surveys
8. In the school teacher example in this unit, what was being tested?
- A) Whether smaller class sizes improve test scores
 - B) Whether providing students with printed notes helps improve their exam performance**
 - C) Whether longer school days improve grades
 - D) Whether online classes are more effective than in-person classes
-

Topic 5.2.2 & 5.2.3: Data Preparation, Cleaning, and Transformation

1. Data preparation includes:
- A) Only collecting new data
 - B) Cleaning data to remove errors, organizing it meaningfully, and converting it into a suitable format for analysis**
 - C) Only visualizing data using charts
 - D) Only calculating the mean and median

2. Data cleaning is best defined as:
- A) Creating new visualizations from clean data
 - B) Correcting or removing problems in the data, such as incorrect entries, missing values, or duplicate results**
 - C) Collecting new survey responses
 - D) Calculating variance and standard deviation
3. In the student score example (Tables 5.1 and 5.2) in this unit, which issues were identified in the original data?
- A) Only missing scores
 - B) Incorrect names, missing scores, and duplicate entries**
 - C) Only duplicate entries
 - D) Incorrect class sections only
4. What happens if data cleaning errors are not fixed before analysis, according to this unit?
- A) The analysis will always be more accurate
 - B) The results of the analysis will be misleading**
 - C) The data automatically corrects itself
 - D) Nothing — errors have no effect on analysis
5. Data transformation is best described as:
- A) Correcting spelling errors only
 - B) Converting data into a format that is easier to work with, such as creating new columns or aggregating data**
 - C) Collecting brand-new survey data
 - D) Removing all rows with missing values
6. Which of the following is an example of data transformation mentioned in this unit?
- A) Fixing a misspelled student name
 - B) Aggregating individual student grades into class-level statistics, such as average scores**
 - C) Conducting a new survey
 - D) Flipping a coin to generate probability data
-

Topic 5.2.3.3: Handling Missing Data

1. Which of the following is NOT one of the three strategies for handling missing data mentioned in this unit?
- A) Imputation
 - B) Flagging
 - C) Removal
 - D) Duplication**

2. Imputation, as a strategy for handling missing data, involves:
 - A) Deleting the row with missing data entirely
 - B) Estimating the missing value using existing data, such as assigning a class average**
 - C) Ignoring the missing data completely
 - D) Creating a new visualization to represent the missing value
 3. Flagging, as a strategy for handling missing data, involves:
 - A) Estimating a value to replace the missing data
 - B) Adding a note in the dataset indicating that a value is missing, ensuring transparency**
 - C) Removing the entire dataset
 - D) Converting missing values into the mean automatically
 4. Removal, as a strategy for handling missing data, involves:
 - A) Estimating the missing value using the class average
 - B) Adding a note to flag the missing value
 - C) Excluding the record with missing data from specific analyses**
 - D) Automatically correcting the missing value using AI
 5. What is a key risk associated with the "Removal" strategy for handling missing data, according to this unit?
 - A) It always introduces bias into the results
 - B) It risks losing valuable information about the excluded record**
 - C) It is illegal in most data analysis contexts
 - D) It requires more computing power than other methods
-

Topic 5.3.1: Introduction to Statistical Modeling and Model Development

1. Statistical modeling is best described as:
 - A) A method of creating bar charts and histograms
 - B) An approach used to analyze data, make sense of the real world, and predict future outcomes**
 - C) A method exclusively for collecting survey data
 - D) A method for encrypting sensitive data
2. According to this unit, what is the first step in building a statistical model?
 - A) Collect data
 - B) Define the problem**
 - C) Choose an algorithm
 - D) Evaluate the model

3. According to this unit, what is the final step in building a statistical model?
- A) Define the problem
 - B) Collect data
 - C) Train the model
 - D) Evaluate the model**
4. What does "training the model" mean, according to this unit?
- A) Testing the model using new, unseen data
 - B) The model learns from the collected data to make predictions**
 - C) Identifying the problem to be solved
 - D) Creating a visualization of the results
5. Why is the "Evaluate the Model" step considered important, according to this unit?
- A) It is optional and can be skipped
 - B) It ensures the model makes accurate predictions by testing it with new or unseen data**
 - C) It is only necessary for logistic regression models
 - D) It replaces the need for data collection
-

Topic 5.3.1.2: Linear Regression

1. Linear regression is best described as:
- A) A method for grouping similar data points together
 - B) A statistical model that helps understand the relationship between two variables, often used to predict one variable based on another**
 - C) A method exclusively for categorical yes/no predictions
 - D) A type of data visualization
2. In the fruit stall example in this unit, what is the independent variable?
- A) Daily earnings
 - B) The number of customers who visit the stall**
 - C) The price of fruit
 - D) The day of the week
3. In the fruit stall example in this unit, what is the dependent variable?
- A) The number of customers
 - B) Daily earnings**
 - C) The type of fruit sold
 - D) The location of the stall

4. In the simple linear regression formula $Y = \beta_0 + \beta_1 X + \varepsilon$, what does β_0 represent?
- A) The dependent variable
 - B) The intercept — the value of Y when X = 0**
 - C) The error term
 - D) The independent variable
5. In the simple linear regression formula $Y = \beta_0 + \beta_1 X + \varepsilon$, what does β_1 represent?
- A) The intercept
 - B) The slope, showing how much Y changes with each unit change in X**
 - C) The error term
 - D) The dependent variable
6. In the simple linear regression formula $Y = \beta_0 + \beta_1 X + \varepsilon$, what does ε represent?
- A) The intercept
 - B) The slope
 - C) The error term, accounting for the difference between predicted and actual values**
 - D) The independent variable
7. Based on the fruit stall example in this unit, what is the calculated slope (β_1)?
- A) 10
 - B) 40**
 - C) 100
 - D) 200
8. Based on the fruit stall example in this unit, what is the calculated intercept (β_0)?
- A) 40
 - B) 100**
 - C) 500
 - D) 980
9. Using the final equation $\text{Earnings} = 100 + 40 \times \text{Customers}$, what would the predicted earnings be for 22 customers?
- A) 880 Rs
 - B) 980 Rs**
 - C) 1,100 Rs
 - D) 1,220 Rs
10. In the fruit stall example, on the 6th day, 28 customers visited and the stall owner earned 1,250 Rs. What was the predicted earnings using the model?
- A) 980 Rs
 - B) 1,220 Rs**
 - C) 1,250 Rs
 - D) 1,300 Rs

11. In the fruit stall example, what was the calculated error on the 6th day (predicted minus actual earnings)?
- A) +30 Rs
 - B) -30 Rs**
 - C) 0 Rs
 - D) -1,220 Rs
12. Which of the following is a suggested tip for improving a statistical model, according to this unit?
- A) Use fewer data points to simplify calculations
 - B) Use more data points for better accuracy and regularly update the model with new data**
 - C) Avoid testing predictions against actual results
 - D) Ignore any relevant external factors
-

Topic 5.3.1.3: Logistic Regression

1. Logistic regression is best used when we want to:
- A) Predict an exact numerical score
 - B) Predict an outcome that can be categorized as "yes" or "no"**
 - C) Group similar data points together
 - D) Calculate the mean of a dataset
2. In the exam example in this unit, what outcome does logistic regression help predict?
- A) The exact exam score a student will receive
 - B) The probability that a student will pass or fail based on hours studied**
 - C) The number of students in a class
 - D) The average score of the entire class
3. How does logistic regression differ from linear regression, according to this unit?
- A) Logistic regression predicts exact numerical values, while linear regression does not
 - B) Logistic regression provides a probability value between 0 and 1, rather than predicting exact numbers**
 - C) Logistic regression cannot be used for real-world predictions
 - D) Logistic regression does not require any data collection
-

Topic 5.3.1.4: Clustering Techniques

1. Clustering is best described as:
 - A) Predicting a "yes" or "no" outcome
 - B) A way of grouping similar things together based on their characteristics**
 - C) Calculating the mean and mode of a dataset
 - D) A method for handling missing data
 2. In the student clustering example in this unit, which two subjects were used to group students?
 - A) Science and History
 - B) Math and English**
 - C) Art and Music
 - D) Physics and Chemistry
 3. K-means clustering is best described as:
 - A) A method for predicting exact numerical outcomes
 - B) One of the simplest and most popular techniques for grouping data into a chosen number of clusters based on similarity**
 - C) A method exclusively for handling missing data
 - D) A type of data visualization tool
 4. In K-means clustering, what does "K" represent?
 - A) The total number of features in the dataset
 - B) The number of clusters to be formed**
 - C) The number of data points in the dataset
 - D) The number of iterations required for convergence
 5. In the student clustering example, which two students were grouped together as strong in math?
 - A) Anie and Tallat
 - B) Basim and Umer**
 - C) Maliha and Anie
 - D) Tallat and Maliha
-

Topic 5.3.2: Evaluating and Interpreting Models

1. Model evaluation is best described as:
 - A) The process of collecting new survey data
 - B) Checking how well a model performs and understanding the results it provides**
 - C) The process of visualizing raw data
 - D) The process of removing missing values

2. Error metrics measure:
- A) The percentage of correct predictions made by a model
 - B) How much the model's predictions differ from the actual values**
 - C) The number of clusters formed by a model
 - D) The slope of a linear regression line
3. Accuracy metrics measure:
- A) How much a model's predictions differ from actual values
 - B) The percentage of correct predictions made by a model**
 - C) The intercept of a regression model
 - D) The number of missing values in a dataset
4. Interpreting a model's output means:
- A) Ignoring the results and starting over
 - B) Understanding what the results reveal, such as drawing conclusions about relationships between variables**
 - C) Only checking for spelling errors in the data
 - D) Only calculating the mean of the predictions
5. Which of the following is an ethical consideration discussed in this unit regarding statistical models?
- A) Model speed only
 - B) Fairness, bias, and data privacy**
 - C) The color scheme of the model's visualizations
 - D) The programming language used to build the model
6. Why is fairness and bias an important ethical consideration in model building, according to this unit?
- A) It has no real-world impact
 - B) A model, such as one used to decide loan approvals, should not unfairly favor one group of people over another**
 - C) It only applies to logistic regression models
 - D) It only matters for models used in sports
-

Topic 5.4: Introduction to Data Visualization and Types of Visualizations

1. Data visualization is best defined as:
- A) The process of collecting raw survey data
 - B) The process of representing data in a visual format, such as graphs or charts, to help identify patterns, trends, and insights**
 - C) The process of calculating variance and standard deviation
 - D) A method for handling missing data

2. Bar charts are best suited for:
- A) Showing trends over time
 - B) Comparing different categories, where the height of each bar indicates its value**
 - C) Displaying the relationship between two continuous variables
 - D) Summarizing data distribution using quartiles
3. Line graphs are best suited for:
- A) Comparing distinct categories
 - B) Showing trends over time by plotting data points connected with a line**
 - C) Displaying outliers using quartiles
 - D) Grouping data into bins or intervals
4. Histograms are best suited for:
- A) Comparing sales figures between products
 - B) Showing the distribution of a dataset by grouping data into bins or intervals**
 - C) Showing changes in temperature over a week
 - D) Displaying only categorical data
5. Scatterplots are best suited for:
- A) Comparing categories using bars
 - B) Showing trends over time
 - C) Displaying the relationship between two variables, where each point represents an observation**
 - D) Summarizing data using medians and quartiles
6. Boxplots (or whisker plots) are best suited for:
- A) Showing exact daily earnings over time
 - B) Summarizing data distribution by displaying the median, quartiles, and potential outliers**
 - C) Comparing only two data points at a time
 - D) Displaying raw, unorganized survey responses
7. In the example given in this unit, which visualization type would best show the relationship between hours studied and exam scores?
- A) Bar chart
 - B) Line graph
 - C) Scatterplot**
 - D) Boxplot
8. In the example given in this unit, which visualization type would best compare the exam score performance of three different classes?
- A) Histogram
 - B) Line graph
 - C) Boxplot**
 - D) Scatterplot

9. In the example given in this unit, which visualization type would best show how temperature changes throughout a single day?
- A) Bar chart
 - B) Line graph**
 - C) Boxplot
 - D) Scatterplot
-

Topic 5.5: Tools for Data Visualization

1. Which tools are specifically mentioned in this unit as easy-to-use options for creating data visualizations?
 - A) Only Python and Tableau
 - B) Microsoft Excel and Google Sheets**
 - C) Only Matplotlib
 - D) Only specialized programming languages
 2. According to the step-by-step guide in this unit, what is the first step in creating a visualization in Excel or Google Sheets?
 - A) Choose a chart type
 - B) Enter your data into the spreadsheet**
 - C) Customize the chart with labels
 - D) Select the data range
 3. According to the step-by-step guide in this unit, which tab is used to insert a chart in Excel or Google Sheets?
 - A) The "Data" tab
 - B) The "Insert" tab**
 - C) The "Format" tab
 - D) The "Review" tab
 4. Why is customizing a chart with labels (such as axis titles) considered an important step, according to this unit?
 - A) It is not important and can be skipped
 - B) It makes the chart clearer and easier to interpret**
 - C) It automatically corrects errors in the underlying data
 - D) It changes the underlying data values
-

Section B: Short Answer Questions (SQs)

Topic 5.1: Basic Statistical Concepts (Overview)

1. Define data analytics in your own words.
2. Define statistics and explain briefly why it is useful for analyzing large datasets.

Topic 5.1.1: Measures of Central Tendency

1. Define mean, median, and mode.
2. Differentiate between how the median is calculated for an odd-numbered dataset versus an even-numbered dataset.
3. Explain briefly what it means for a dataset to have more than one mode.
4. Predict the mean of the dataset: 40, 50, 60, 70, 80.
5. Predict the median of the dataset: 15, 20, 25, 30.

Topic 5.1.2: Measures of Dispersion (Variance and Standard Deviation)

1. Define variance and explain what a higher variance indicates about a dataset.
2. Define standard deviation and explain briefly how it relates to variance.
3. Differentiate between variance and standard deviation in terms of interpretability.
4. Explain briefly why Class B (from this unit's worked example) is considered more spread out than Class A.
5. State the formula used to calculate variance, identifying what each symbol represents.

Topic 5.1.3: Introduction to Probability

1. Define probability in your own words.
2. State the formula used to calculate probability.
3. Explain briefly why the probability of flipping a coin and getting heads is 50%.
4. List two real-world areas, other than coin flips, where probability is used, according to this unit.

Topic 5.2.1: Data Collection Methods

1. Define data collection and explain why it is an important first step in research.
2. Differentiate between surveys and observations as data collection methods.
3. Explain briefly how experiments differ from surveys and observations.
4. Identify which data collection method was used in the grocery store example, and briefly explain why it was appropriate.
5. Identify which data collection method was used in the restaurant seating example, and briefly explain why it was appropriate.

Topic 5.2.2 & 5.2.3: Data Preparation, Cleaning, and Transformation

1. Define data preparation and list its main components.
2. Define data cleaning and explain briefly why it is necessary before analysis.
3. Differentiate between data cleaning and data transformation.
4. Explain briefly, using the student score example, what kinds of errors data cleaning might fix.
5. Explain briefly how data transformation might be used to summarize individual student grades at the class level.

Topic 5.2.3.3: Handling Missing Data

1. List the three strategies for handling missing data discussed in this unit.
2. Differentiate between imputation and flagging as strategies for handling missing data.
3. Explain briefly one risk associated with removing records that have missing data.
4. Using the Sara example from this unit, briefly explain how imputation could be applied to estimate her missing score.

Topic 5.3.1: Introduction to Statistical Modeling and Model Development

1. Define statistical modeling in your own words.
2. List the five steps involved in building a statistical model, in the correct order.
3. Explain briefly what happens during the "Train the Model" step.
4. Explain briefly why the "Evaluate the Model" step is considered important.

Topic 5.3.1.2: Linear Regression

1. Define linear regression and explain what kind of relationship it helps us understand.
2. Differentiate between the independent variable and the dependent variable, using the fruit stall example.
3. State the simple linear regression formula and briefly explain what each symbol (Y , X , β_0 , β_1 , ϵ) represents.
4. Explain briefly what the slope (β_1) represents in a linear regression model.
5. Explain briefly what the intercept (β_0) represents in a linear regression model.
6. Predict the output of the model $\text{Earnings} = 100 + 40 \times \text{Customers}$, if 15 customers visit the stall.
7. Explain briefly what "error" means in the context of evaluating a linear regression model's predictions.

Topic 5.3.1.3: Logistic Regression

1. Define logistic regression in your own words.

2. Differentiate between linear regression and logistic regression.
3. Explain briefly what kind of output logistic regression provides.

Topic 5.3.1.4: Clustering Techniques

1. Define clustering in your own words.
2. Explain briefly how K-means clustering groups data together.
3. Identify what the letter "K" represents in K-means clustering.
4. Using the student example in this unit, briefly explain how Basim and Umer might be grouped into the same cluster.

Topic 5.3.2: Evaluating and Interpreting Models

1. Define model evaluation in your own words.
2. Differentiate between error metrics and accuracy metrics.
3. Explain briefly what it means to "interpret" a model's output.
4. List two ethical considerations discussed in this unit regarding statistical models.
5. Explain briefly why fairness and bias matter when a model is used to make decisions, such as loan approvals.

Topic 5.4: Introduction to Data Visualization and Types of Visualizations

1. Define data visualization and explain briefly why it is useful.
2. Differentiate between a bar chart and a line graph, including when each should be used.
3. Explain briefly what a histogram shows, and how it differs from a bar chart.
4. Explain briefly what a scatterplot displays.
5. Explain briefly what a boxplot displays, including the role of quartiles and outliers.
6. Identify which visualization type would be most appropriate for comparing three classes' exam score distributions.

Topic 5.5: Tools for Data Visualization

1. List two tools mentioned in this unit for creating data visualizations.
 2. State the first and second steps in the step-by-step guide for creating a chart in Excel or Google Sheets.
 3. Explain briefly why customizing a chart with axis labels is an important step.
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Section C: Long / Extensive Questions (LQs)

Topic 5.1.1 & 5.1.2: Measures of Central Tendency and Dispersion

1. Discuss the concept of measures of central tendency with examples.
 - a) Define mean, median, and mode, and explain how each is calculated.
 - b) Using a dataset of your own choice (at least 6 values), calculate the mean, median, and mode, showing all working steps.
 - c) Evaluate a scenario where the mean might be misleading as a measure of central tendency (for example, a dataset with an extreme outlier), and justify why the median might be more appropriate in that case.
2. Analyze the concept of measures of dispersion — variance and standard deviation — using detailed calculations.
 - a) Explain the formulas for variance and standard deviation, defining every symbol used.
 - b) Reconstruct the full step-by-step variance and standard deviation calculations for Class A (50, 52, 55, 57, 60) and Class B (30, 45, 55, 75, 90), as presented in this unit, showing the mean, squared deviations, variance, and standard deviation for each class.
 - c) Evaluate what the difference in standard deviation between Class A and Class B reveals about the two classes' score distributions, and discuss a real-world scenario where understanding this difference would matter (e.g., for a teacher planning lessons).

Topic 5.1.3: Introduction to Probability

1. Discuss the concept and real-world relevance of probability.
 - a) Define probability and state its formula, explaining each component.
 - b) Construct a probability calculation for a scenario of your own choice (different from the coin-flip example), showing the number of favorable outcomes and total outcomes.
 - c) Evaluate at least two real-world fields (such as weather forecasting or sports) where probability plays an important decision-making role, explaining how probability might be applied in each.

Topic 5.2.1: Data Collection Methods

1. Explain data collection methods in comprehensive detail.
 - a) Define surveys, observations, and experiments, and describe a real-world example of each, referencing the examples from this unit.
 - b) Construct a comparison table evaluating surveys, observations, and experiments based on the type of data they collect, the level of researcher involvement, and typical use cases.
 - c) Design a new data collection scenario (different from those already discussed) for a school library wanting to understand student reading habits, and justify which of the three methods would be most appropriate, explaining your reasoning.

Topic 5.2.2 & 5.2.3: Data Preparation, Cleaning, Transformation, and Missing Data

1. Discuss in detail the process of preparing raw data for analysis.
 - a) Define data cleaning and data transformation, and explain how each contributes to reliable analysis results.
 - b) Using the student score example (Tables 5.1 and 5.2) from this unit, reconstruct the data cleaning process, identifying each error and explaining how it was corrected.
 - c) Evaluate the three strategies for handling missing data — imputation, flagging, and removal — discussing a scenario where each strategy would be the most appropriate choice, and the risks associated with choosing incorrectly.

Topic 5.3.1: Statistical Modeling and Model Development

1. Explain the role and importance of statistical models in solving real-world problems.
 - a) Define statistical modeling and describe the five steps involved in building a statistical model, in order.
 - b) Discuss a real-world scenario (such as predicting monthly grocery expenses) and describe how each of the five steps would be applied to that scenario.
 - c) Evaluate why testing a model with new, unseen data (Step 5) is considered essential, and discuss what risks arise if this step is skipped.

Topic 5.3.1.2: Linear Regression

1. Describe the steps involved in building a basic statistical model (e.g., linear regression). Include details on data collection, model training, and evaluation.
 - a) Reconstruct the full fruit stall linear regression example from this unit, showing how the slope ($\beta_1 = 40$) and intercept ($\beta_0 = 100$) were calculated from the given data.
 - b) Using the final equation $\text{Earnings} = 100 + 40 \times \text{Customers}$, calculate the predicted earnings for three different customer counts of your choice, showing your work.
 - c) Discuss the concept of "error" in model evaluation, using the 6th-day example from this unit (28 customers, actual earnings of 1,250 Rs, predicted earnings of 1,220 Rs), and evaluate what this error reveals about the limitations of real-world predictive models.

Topic 5.3.1.3 & 5.3.1.4: Logistic Regression and Clustering Techniques

1. Analyze and compare logistic regression and clustering as two different statistical modeling techniques.
 - a) Define logistic regression and explain, using the exam pass/fail example from this unit, how it differs from linear regression.
 - b) Define clustering and K-means clustering, and reconstruct the student clustering example from this unit (Basim, Umer, Anie, Tallat, and Maliha), explaining how the algorithm would group students based on math and English scores.

- c) Evaluate a new real-world scenario where logistic regression would be more appropriate than clustering, and a separate scenario where clustering would be more appropriate than logistic regression, justifying each choice.

Topic 5.3.2: Evaluating and Interpreting Models

1. Discuss in detail how statistical models are evaluated and interpreted, including their ethical implications.
 - a) Differentiate between error metrics and accuracy metrics, providing an example of each based on this unit's content.
 - b) Explain what it means to interpret a model's output, using the example of a linear regression model showing that study hours strongly affect exam scores.
 - c) Evaluate the ethical considerations of fairness, bias, and data privacy in model building, discussing a real-world scenario (such as a loan approval system) where failing to consider these issues could cause harm.

Topic 5.4: Types of Data Visualizations

1. Discuss the types of data visualizations and their uses.
 - a) Define bar charts, line graphs, histograms, scatterplots, and boxplots, describing what each is best suited to display.
 - b) Construct a table matching each visualization type to a specific real-world scenario from this unit (such as comparing product sales, tracking temperature, or comparing class performance).
 - c) Evaluate a new dataset scenario of your own choice, and justify which visualization type would be most appropriate for representing it, explaining your reasoning.

Topic 5.5: Tools for Data Visualization

1. Explain, in detail, how to create and interpret data visualizations using tools such as Excel and Google Sheets.
 - a) Describe the full step-by-step process for creating a chart in Excel or Google Sheets, as outlined in this unit.
 - b) Discuss why entering data correctly and selecting the appropriate chart type are both essential steps in producing a clear, accurate visualization.
 - c) Evaluate the benefits and limitations of using tools like Excel and Google Sheets for data visualization, compared to more advanced tools such as Python, Tableau, or Matplotlib (mentioned in the unit's learning outcomes).